
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Generative image models trained on large-scale datasets achieve strong aggregate performance, yet they can still miss visually simple concepts—e.g., human hands or objects appearing in groups of four—that should be represented in the training distribution. Because these failures have mostly been reported as isolated anecdotes, it remains unclear whether they are rare exceptions or measurable, model-wide discrepancies in concept prevalence. We address this gap with a systematic framework for identifying and quantifying “conceptual blindspots”: concepts present in natural data whose occurrence is suppressed or exaggerated in model generations. Our method uses sparse autoencoders (SAEs) to extract interpretable concept embeddings and directly compare real-versus-generated concept prevalence under a shared metric. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—which provides the resolution needed to measure fine-grained empirical margins between natural and synthesized image distributions. Across four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the same comparison protocol surfaces both suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual-datapoint level, the metric further separates high-alignment cases from memorization artifacts in which models reproduce highly specific visual templates seen during training. Overall, we provide a theoretically grounded and empirically operational framework for measuring conceptual fidelity relative to the data-generating process, making blindspot comparisons explicit across models, concepts, and evaluation conditions.

1 Introduction

Generative image models trained on large-scale datasets now support a wide range of applications, both within vision [1, 9] and beyond it [9, 15]. Their strong headline performance, however, masks measurable failures on concepts that are visually simple and plausibly frequent in the data, including human hands [16–19], objects appearing in groups of four [20], and negations or object relations [21, 22]; these patterns appear in both qualitative reports [23–25] and quantitative studies [22, 26]. When prompted for these concepts, models often generate nearby visual structures without matching the target concept itself—for example, a hand-like object with six fingers rather than a correct human hand. We call these failures “conceptual blindspots”¹ cases where the generated distribution assigns too little or too much mass to a concept despite apparent exposure to demonstrations of that concept. The central empirical question is therefore comparative: do such errors remain isolated quirks of a few prompts, or do they form systematic, quantifiable deviations in concept prevalence across a broad concept inventory and multiple model families?

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1: $\delta(k)$ measures the real-versus-generated prevalence margin for concept c_k relative to its natural-data frequency. Concepts with $\delta(k) < 0.1$ are counted as **suppressed conceptual blindspots**, whereas concepts with $\delta(k) > 0.9$ are counted as **exaggerated conceptual blindspots**. The examples compare four popular generative image models under the same prompt-conditioned evaluation, showing both blindspot cases and aligned concepts.

Existing evaluation protocols make this comparison difficult at the required granularity [28, 29]. Common aggregate metrics such as FID [30] summarize image realism and distributional similarity, but they do not report which concepts account for the observed gap. Metrics such as CLIPScore, generation-diversity measures, and distribution-coverage statistics [31–33] provide useful global signals, yet they still operate above the level of fine-grained features or concepts [34, 35]; as a result, they do not expose the per-concept margins needed to identify blindspots. Qualitative evaluations, including participant surveys [5, 24, 36, 38] and open-ended exploration [39], can reveal salient failures, but their evidence is harder to scale, repeat, and compare consistently across model families.

This work. We therefore frame conceptual blindspot discovery as an automated, unsupervised measurement problem: for each concept in the data distribution, estimate whether its generated prevalence differs from its prevalence under the data-generating process, and make those differences comparable across models. Our contributions in this work are three-fold:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We introduce a framework that defines blindspots as measurable concept-prevalence deviations between generated and natural images (Section 3). This turns anecdotal failures into a quantitative comparison problem with explicit suppressed and exaggerated regimes.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We develop a structured SAE-based pipeline that extracts a shared concept basis and compares natural versus synthesized concept distributions under the same metric (Section 4). To support fine-grained measurement, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts, the largest such RA-SAE to date.
- **Exploratory Tool and Analysis.** Our exploratory web tool supports distribution-level, datapoint-level, and cross-model comparisons of blindspots (Sections 5.1–5.4). Applying the same evaluation to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5), we identify suppressed blindspots such as bird feeders, DVD discs, and whitespaces on documents, as well as exaggerated blindspots such as wood background texture and palm trees (Fig. 1).

The model, code, and web tool are available at a software repository.

2 Related Work

2.1 Explainability in Vision

Early work in explainable AI for vision primarily attributed predictions to influential input regions [40–42]. These attribution maps provide local evidence but limited semantic accounting of the representation, and they can produce unreliable explanations [43–45]. Concept-based interpretability [46] shifted the comparison unit from pixels to semantically meaningful directions, enabling analyses of which concepts and structures models use rather than only where they attend [47–49].

Recent work shows that ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51] can be viewed as dictionary-learning methods with different constraints [52]. Among them, SAEs are especially useful for quantitative concept analysis because they scale to large dictionaries and return sparse activation values that can be compared across datasets. Although original SAEs can produce overly specific features [53], struggle with compositionality [54], and yield limited intervention

76 effects [55], improved variants such as archetypal SAE (RA-SAE) [56] provide a more stable basis
77 for the concept-level comparisons used here.

78 Other interpretability routes—including prompt-based probing [57], attention or activation visual-
79 izations [58, 59], and dataset-level statistics [32, 33] such as diversity or coverage metrics—provide
80 complementary evidence but only partially answer our question. Their comparison units are typically
81 full prompts, images, or aggregate distributions rather than fine-grained concepts [34, 35]. They also
82 often require subjective interpretation and do not cleanly separate different failure modes [60]. Thus,
83 existing methods are not designed to produce the concept-by-concept empirical margins needed to
84 identify feature- and concept-level weaknesses in generative image models [28].

85 2.2 Generative Image Models

86 Diffusion-based methods now dominate generative vision across images [4, 61, 64], video [65, 72],
87 and 3D [7, 73, 75]. For image generation, this trajectory begins with denoising diffusion probabilistic
88 models (DDPMs) [76] and extends to non-Markov diffusion processes such as denoising diffusion
89 implicit models (DDIMs) [77]. These model families make cross-architecture empirical comparison
90 especially important because strong sample quality can coexist with different concept-level error
91 profiles.

92 The Stable Diffusion (SD) [9] family made DDPM-based text-to-image generation broadly accessible
93 in research and open-source settings. Subsequent versions, including SD 2 [78], SD 3 [79], and SD
94 XL [80], as well as extensions for style [81, 82], pose [83], identity [84, 85], and image-to-image
95 inputs, create a diverse evaluation landscape. Related latent diffusion systems such as Kandinsky [86],
96 PixArt [87], DeepFloyd [4, 88], and FLUX [89] further motivate a metric that can compare concept
97 fidelity across architectures rather than treating each model in isolation.

98 2.3 Datasets for Generative Image Models

99 The empirical performance of modern generative vision models is tightly linked to large-scale
100 compute and internet-scale datasets [90, 92]. LAION-5B [93], scraped from Common Crawl [94], is
101 especially central for open-source text-to-image models such as SD and its derivatives: it contains
102 over 5B image-caption pairs, including 2.3B English pairs. Other important training resources include
103 COYO-700M [95] with 700M pairs and Conceptual Captions [96] with 3M pairs, underscoring that
104 blindspot analysis must reason about empirical frequency and long-tail coverage rather than only
105 model architecture.

106 As LAION-5B became widely used, empirical audits also documented substantial dataset biases
107 and harmful content [97, 101]. Filtering did not remove all problematic material [97, 99], including
108 NSFW content [100] and hundreds of CSAM instances [101], which led to temporary removal from
109 official channels. In addition, low-quality images [102] and internet-style captions such as product
110 descriptions can diverge from user prompting behavior [103], making dataset-model comparisons an
111 important empirical baseline for interpreting generated concepts.

112 2.4 Concept Discovery and Sparse Coding in Generative Image Models

113 Dictionary learning provides the quantitative machinery for representing each input as a sparse
114 combination of a small number of atoms [104, 106]. Building on compressed sensing theory [107,
115 108], the field has progressed from vector quantization [109] to methods such as Non-negative Matrix
116 Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. Online algorithms [114], structured
117 sparsity [115], theoretical guarantees [116, 117], and links to deep learning [118, 119] make sparse
118 coding a natural basis for comparing concept activations across real and generated datasets.

119 Sparse coding has also been used to study high-level concepts inside diffusion models [120]. Earlier
120 concept-grounded approaches for generative models include concept-bottleneck models, which
121 require human intervention during training [121], and post-hoc detectors that add concept supervision
122 after training [122]. Because both rely on human-defined concept sets, they can measure known
123 targets but may miss broader empirical trends that are not specified in advance.

124 Aggregate metrics such as precision, recall, density, and coverage [35, 123], along with latent density
125 scores that estimate sample quality from the model’s latent space [124], are valuable for detecting
126 distributional gaps. Their limitation for our purpose is not sensitivity but resolution: they usually

do not identify which specific concepts are under- or over-represented, nor do they quantify those concept-level margins.

Together, these limitations motivate a scalable, unsupervised framework that reports concept-level failure modes and their magnitudes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We first formalize *conceptual blindspots* as systematic discrepancies between the concept distribution in natural images and the concept distribution induced by a generative model. The goal of the formalization is to turn each concept into a comparable empirical quantity: under the same prompts, is the concept less prevalent, similarly prevalent, or more prevalent in generated images than in natural data? The comparison pipeline is illustrated in Fig. 2. Although the derivation uses standard assumptions [126, 130], our experiments show that the resulting metric still exposes meaningful empirical structure when these assumptions are only approximate.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational simplicity, we use $p(\cdot)$ for both the latent density $p(c)$ and its pushforward $p(x)$ in image space, where $x = G(c)$. The invertibility of G justifies this shared notation because it induces a valid distribution over $\mathcal{X}$ through the change-of-variables formula. The individual latent dimensions represent the **Concepts** that structure the image distribution $P_{\mathcal{X}}$ over the observation space $\mathcal{X}$. Examples include factors such as color, shape, size, and location [131, 132]; in our empirical analysis, each such factor becomes a dimension on which natural and generated images can be compared. We also allow the data-generating process to associate a text description $t \in \mathcal{T}$ with each sampled image, while leaving the text-generation mechanism implicit. A text-conditioned **Generative image model** g_{θ} , with parameters θ , is then trained on image-text pairs to map a noise signal $\eta \sim \mathcal{N}(0, I)$ and a scene description t to an image x illustrating that description.

To identify blindspots for g_{θ} , we compare the probability mass assigned to each concept by the data-generating process with the mass induced by the model under matched prompts. We therefore start from an evaluation set of natural image-text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ and generate the matched synthetic set $\mathcal{D}'_{\mathcal{X}}$ using the same text descriptions. Concept occurrence is estimated with an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps each image to a d -dimensional concept vector, where the k^{th} entry is the scalar energy of concept k , i.e., $\xi_k(x) = E(c_k)$. The invertibility assumption makes these estimates comparable to the latent data-generating process. Accordingly, the probability assigned to image x can be written as $p_k(x) \propto \exp(-\sum_k \xi_k(x))$ (with Z_{ξ} as the partition function), yielding the population-level estimate $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{x \in \mathcal{D}_{\mathcal{X}}} p_k(x)$. Using this quantity and the sigmoid map $\sigma(\cdot)$, we define the average energy difference for concept k between the natural dataset $\mathcal{D}_{\mathcal{X}}$ and the generated dataset $\mathcal{D}'_{\mathcal{X}}$.

Definition 2 (Energy Difference). Let $x \in \mathcal{D}_{\mathcal{X}}$ denote a real image sampled from the data-generating process G , and let $x' \in \mathcal{D}'_{\mathcal{X}}$ be a synthetic image generated by the model g_{θ} . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the

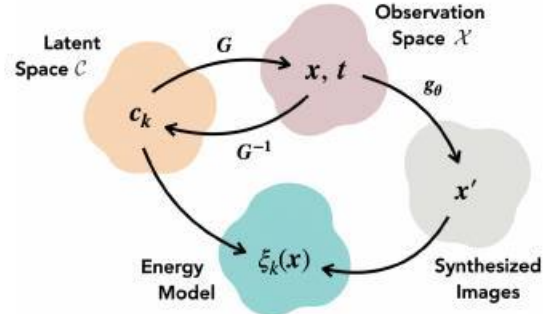


Figure 2: **Intuitive mapping of our framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) through the (invertible) DGP. g_{θ} generates comparison images x' from the same text t . The energy model $\xi_k(x)$ places x and x' in a shared concept space, enabling per-concept prevalence comparisons that identify suppressed and exaggerated blindspots.

179 energy difference for concept k as:

$$\begin{aligned}\delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})},\end{aligned}\quad (1)$$

180 where the expectations are taken over $\mathcal{D}'_{\mathcal{X}}$ and $\mathcal{D}_{\mathcal{X}}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$
181 denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

182 Thus, the energy difference for concept k is a normalized real-versus-generated prevalence ratio: it
183 compares the conceptual probability mass in the generated observations $\mathcal{D}'_{\mathcal{X}}$ to the corresponding
184 baseline mass in the natural observations $\mathcal{D}_{\mathcal{X}}$. This scalar makes suppressed and exaggerated regimes
185 directly comparable across concepts and models.

186 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** Given a generative image model
187 g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in
188 the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

189 In this framing, a conceptual blindspot is a large concept-level prevalence margin between generated
190 images and the data-generating process. Concepts below the lower threshold are treated as suppressed
191 because their generated activation is disproportionately low (e.g., $\delta(k) < \lambda_{\min}$), whereas concepts
192 above the upper threshold are treated as exaggerated because they are overrepresented (e.g., $\delta(k) >$
193 $\lambda_{\max}$). Throughout the empirical analysis, we use $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to isolate these two
194 high-margin regimes.

195 This definition is related to “mode collapse” in prior work (e.g., Bau et al. [39]), but it changes the
196 unit of measurement. Mode collapse concerns the exaggerated or suppressed odds of generating
197 entire images; our metric instead measures the changed odds of individual concepts. For example, if
198 a model repeatedly fails to produce an object with a white background when that concept appears
199 in the natural baseline, the margin is counted as a suppressed conceptual blindspot.

200 4 Method: Operationalizing the Definition of Conceptual Blindspots

201 We now operationalize the formal definition for a generative model g_θ by specifying the matched
202 datasets and concept-energy representation used in the comparisons. Following Sec. 3 the pipeline
203 requires (i) model samples paired with natural images so the same prompts define the evaluation
204 condition, and (ii) an energy model that maps both image sets into a shared concept basis. Below,
205 $\|\cdot\|_F$ denotes the Frobenius norm and $\|\cdot\|_0$ denotes the number of non-zero entries (the ℓ_0
206 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$ denotes element-wise non-negativity. For $n > 0$, we
207 let $[n] := \{1, \dots, n\}$ and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

208 **From Prompts to Latent Representations.** For each blindspot comparison, we pair a natural
209 dataset $\mathcal{D}_{\mathcal{X}}$ of image-caption pairs $(\mathbf{x}, \mathbf{t})$ sampled from the data-generating process G with synthetic
210 counterparts generated by g_θ from the same text descriptions.

211 Given caption $\mathbf{t}$, we synthesize the matched image $\mathbf{x}'$ with a pretrained text-to-image generator
212 $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87].
213 Sampling follows the standard latent-space reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\gamma_t, \mathbf{t}, \mathbf{c}) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

214 where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0
215 is decoded by a pretrained VAE to produce the generated comparison image $\mathbf{x}' = \text{VAE}(\gamma_0)$. For the
216 rest of the paper, the internal sampling details are abstracted away and g_θ is treated as a black box
217 mapping prompts to images, $\mathbf{t} \mapsto \mathbf{x}'$.

218 **Defining the Energy Model.** To obtain the shared concept basis, we build on prior results show-
219 ing that self-supervised learning methods can invert the data-generating process and identify the
220 underlying energy function up to linear transformations [126, 128, 133, 134]. We use DINOv2 [135],
221 a state-of-the-art self-supervised vision model, as the feature extractor for all real and generated
222 images. Because the number of concepts in the DGP is expected to exceed the dimensionality of

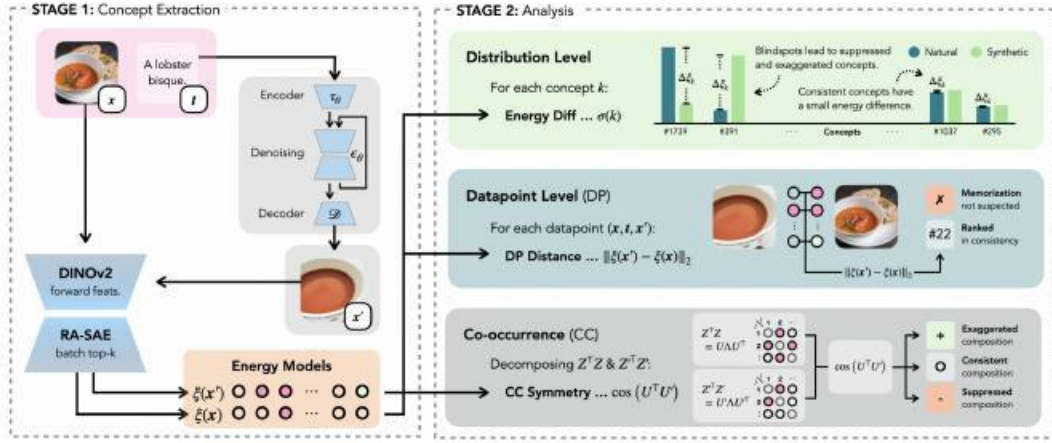


Figure 3: **Concept Extraction Pipeline.** For each matched triplet (x, t, x') , both images are encoded with DINOv2 and projected by RA-SAE into the same sparse 32,000-concept space. The resulting vectors $\xi(x)$ and $\xi(x')$ provide comparable concept energies; in particular, $\xi_k(x) = E(c_k)$ records the energy of the k^{th} concept.

the feature space [51, 136], we train sparse autoencoders (SAEs) on DINOv2 features to recover concept-aligned subspaces [50, 56, 137–139]. If DINOv2 represents the relevant generative concepts as approximately orthogonal directions, consistent with the independence constraint in Def. 1, the SAE can separate those directions into individual latent dimensions [136]. The activation of each SAE dimension then serves as the empirical estimate of the energy assigned to the corresponding concept, enabling direct quantitative comparison between $\mathcal{D}_X$ and $\mathcal{D}'_X$.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the DINOv2 feature extractor. We compute $a = f(x) \in \mathbb{R}^d$ for every natural and synthetic image in $\mathcal{D}_X$ and $\mathcal{D}'_X$. For n image-text pairs, the real and generated features are stacked into matrices $A, A' \in \mathbb{R}^{n \times d}$. Each feature vector is then decomposed into a sparse combination of learned concept atoms using an SAE. Let $D \in \mathbb{R}^{d \times K'}$ be a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the encoder that maps features to sparse codes. Applying Ψ row-wise gives the activation matrix $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, where each row $z_i = \Psi(a_i)$ is the concept decomposition of one image. The decoder reconstructs features as $ZD^\top$, and the SAE objective minimizes reconstruction error subject to sparsity and non-negativity:

$$\min_{\Psi, D} \|A - \Psi(A)D^\top\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs can learn arbitrarily oriented dictionaries, which would make concept-level margins sensitive to random seed rather than model behavior. To improve reproducibility and make comparisons stable, we use the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE anchors the dictionary D as a convex combination of training examples. Specifically, $D = WA$ with $W \in \Omega_{K', n}$, where $\Omega_{K', n}$ is the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1}\}. \quad (3)$$

Thus every atom D_i lies in the convex hull of the data $\text{conv}(A)$, and any reconstruction $ZD^\top$ remains inside the conic hull $\text{cone}(A)$. This constraint keeps learned concepts tied to the empirical support of the data distribution [56], which is essential when interpreting quantitative margins between natural and generated images. After training, the SAE returns comparable sparse codes: Z for real images and Z' for generated counterparts. *These codes represent the same prompt-conditioned visual semantics in a shared, interpretable concept basis, with each activation value serving as the energy used in our blindspot analysis.* Consequently, for every prompt-image pair (t, x) , the pipeline yields two sparse vectors (z, z') whose coordinate-wise differences quantify how the generated image shifts concept energy relative to the natural image.

In full, we start from a captioned image (t, x) , synthesize the matched counterpart x' , and encode both images with DINOv2 and RA-SAE into the shared sparse representations (z, z') . These matched representations are the empirical basis for all distribution-level, model-level, and datapoint-level concept comparisons in the results.

5 Results

We evaluate four LAION-5B-trained generative image models—SD 1.5, SD 2.1, PixArt, and Kandinsky—on the same $|\mathcal{X}| = 10,000$ image-text pairs and their matched generations (Fig. 4). This fixed evaluation set makes the resulting margins comparable across methods, baselines, and concepts. Our analysis spans three levels (Fig. 3): a **distribution-level** evaluation quantifies suppressed and exaggerated concept-prevalence shifts; a **datapoint-level** analysis identifies high- and low-distance image pairs tied to ambiguity, omission, and memorization; and a **compositional** analysis examines distortions in concept co-occurrence geometry.

To make these quantitative comparisons inspectable, we provide the interactive exploratory tool shown in Fig. 5. The web interface centers on a UMAP projection of concept representations and displays the energy-difference value for each concept, allowing users to compare the same 32,000-concept basis across SD 1.5, SD 2.1, PixArt, and Kandinsky. It is publicly available at a software repository together with pre-computed energy-difference data for all four models. The analyses below use this tool to move from aggregate distributions to individual concepts and datapoints. Its primary functionalities are:

- **Contrast different models and architectures.** For each model, the tool displays all 32,000 RA-SAE concepts in a shared UMAP view, with every point color-coded by its energy difference.
- **Inspect concepts.** Each concept card reports key statistics, representative real and generated images (x, x'), and visualized co-occurrence patterns for the selected comparison.
- **Explore blindspots.** Global rankings order suppressed and exaggerated concepts by their measured margins, making the strongest blindspot candidates easy to inspect.

Rather than exhausting a single case study, we report the main empirical patterns that emerge from these comparisons and leave the tool available for broader, customizable follow-up analysis.

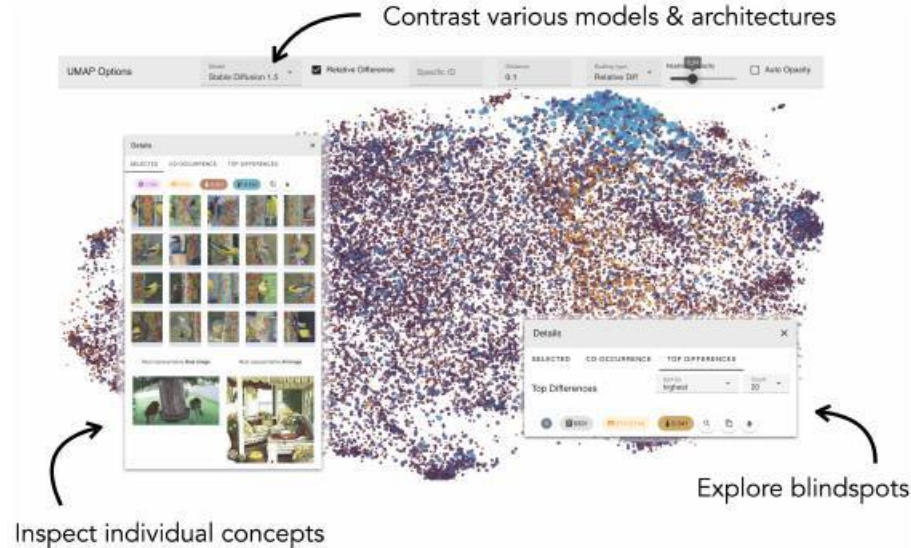


Figure 5: **Overview of the Exploratory Tool.** For each evaluated model, the interface shows a UMAP projection in which each dot is one RA-SAE concept and the color encodes its energy difference. Selecting a concept opens a detail panel with statistics, illustrative natural and generated images x and x' , co-occurrence rankings, and global lists of the strongest suppressed and exaggerated blindspots.

5.1 Generative Image Models Cannot Produce Unseen Concepts

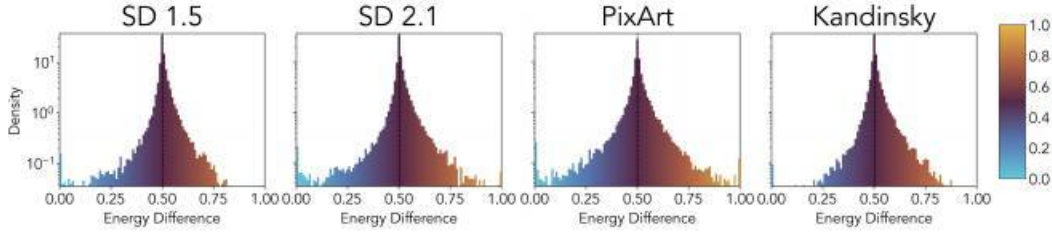


Figure 6: **Concept Energy Distribution.** Log-scale histograms summarize $\delta(k)$ over all 32,000 RA-SAE concepts for each evaluated model, comparing generated concept prevalence against the natural-image baseline. Left-tail values indicate suppressed concepts (under-representation), whereas right-tail values indicate exaggerated concepts (over-representation).

We first quantify distribution-level disparities by computing the marginal energy difference $\delta(k)$ for each of the 32,000 RA-SAE concepts. As defined in Sec. 4, this metric compares the prevalence of concept k in the generated set against its prevalence in the natural-image baseline: $\delta(k) < 0.1$ marks suppression, whereas $\delta(k) > 0.9$ marks exaggeration. Fig. 6 reports the resulting distribution separately for SD 1.5, SD 2.1, PixArt, and Kandinsky. All four histograms are heavy-tailed, with visible mass near both extremes, showing that concept coverage differences are not confined to a few isolated concepts. The left tail, corresponding to suppressed concepts, is consistently denser and longer than the right tail, indicating a recurring empirical tendency toward omission. This left-tail asymmetry is captured by negative skewness in every model: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

The same plots also show that the aggregate pattern is not identical across architectures. All models contain both suppressed and exaggerated concepts, but their spreads and tail shapes differ; PixArt, for example, exhibits a wider spread consistent with a more suppressed concept distribution. Thus, the shared negative skew establishes a cross-model omission pattern, while the differences in spread motivate the more structured comparisons in the following sections.

5.2 Structure and Specificity of Conceptual Blindspots

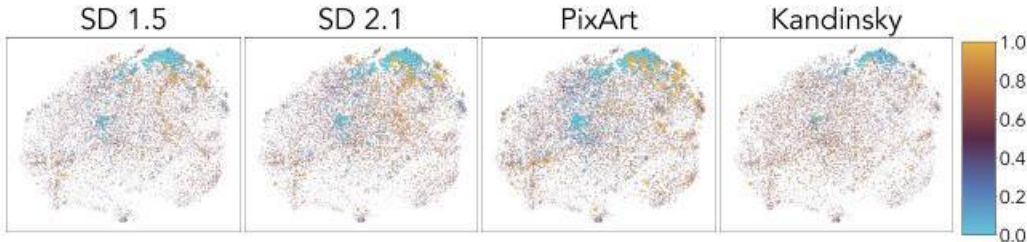


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations place all 32,000 concepts in a shared geometry and color each concept by $\delta(k)$ for the corresponding model. Spatially coherent blue regions indicate clusters of suppressed concepts, while yellow regions indicate clusters of exaggerated concepts.

Having established marginal concept-prevalence gaps, we next ask whether those gaps are randomly scattered or organized in the learned concept space. We embed all 32,000 sparse concept codes with UMAP and color each concept by its model-specific $\delta(\cdot)$ value. Fig. 7 shows that blindspots form distinct clusters across all evaluated models. The most coherent regions often correspond to suppressed concepts (blue), indicating that omissions are structured in concept space rather than distributed uniformly across the dictionary. We then quantify cross-model consistency by comparing the full $\delta(\cdot)$ vectors between models. Fig. 8 reports pairwise scatter plots and Pearson correlations using SD 1.5 as a reference. SD 1.5 and SD 2.1 show the strongest agreement ($r = 0.82$), consistent with their related architectures and training pipelines. The corresponding correlations with PixArt and Kandinsky are lower— $r = 0.41$ and $r = 0.46$, respectively—showing that different architectures shift emphasis toward different regions of the concept space.

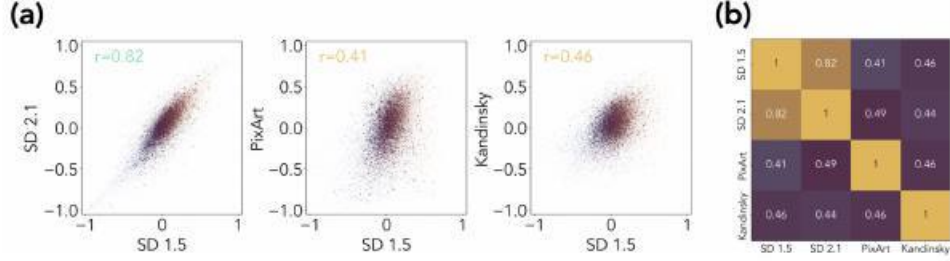


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots compare $\delta(k)$ values across the four evaluated models, with Pearson correlations reported in the upper left of each panel. The high SD 1.5–SD 2.1 correlation contrasts with lower correlations involving PixArt and Kandinsky, separating shared blindspot structure from model-specific deviations. **Correlation Matrix of Conceptual Blindspots.** The heatmap summarizes pairwise Pearson correlations for δ vectors, quantifying how consistently blindspots are shared across models.

Taken together, the cluster and correlation analyses show a two-part empirical pattern: some blindspots are shared across models, consistent with dataset-level effects, while others are model-specific and likely tied to architecture, training dynamics, or capacity. The remaining analyses therefore distinguish universal blindspots from unique ones and examine how both appear in concrete generated images.

5.3 Qualitative Examples of Blindspots

We next ground the quantitative regimes in representative images, using examples whose measured energy differences place them in suppressed or exaggerated regions. Fig. 9a shows a universal suppressed blindspot: all four evaluated models fail to reproduce solid white on documents. The natural image and caption contain the target concept, but the generated images omit the intended visual semantics, matching the distribution-level evidence that this part of concept space is under-sampled. Fig. 9b shows a model-specific suppressed blindspot: pan is captured by three models but is missing from Kandinsky generations. This qualitative contrast mirrors the correlation analysis in Sec. 5.2 where some concept margins are shared while others vary substantially by model.

Fig. 10 presents the complementary high-margin regime: exaggerated generation of shadow under animal. The concept is visually plausible, but the generated images render it more consistently and more strongly than the natural-image baseline, indicating an overactive prior shared across the evaluated models. In our search for model-specific exaggerations, we did not find clear examples, suggesting that the strongest exaggeration patterns are approximately universal in this evaluation.

These qualitative examples show that the measured margins correspond to visible generation behavior: the same diagnostic identifies both shared failures, such as missing whitespaces on documents, and model-specific failures,

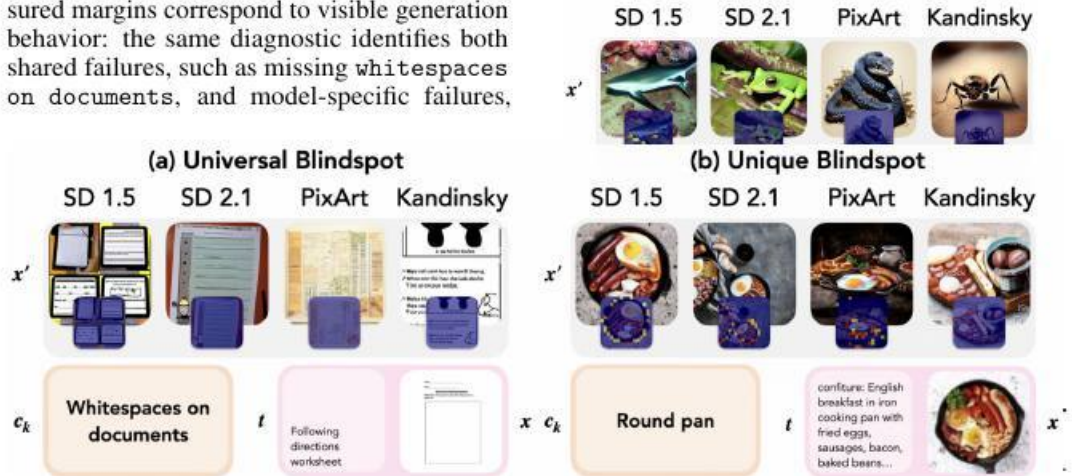


Figure 9: **Examples of Suppressed Conceptual Blindspots.** Natural images back-projected to the corresponding concept c_k , and captions t are shown next to four generated images x' synthesized with S_θ . The first comparison illustrates a universal suppressed blindspot: all four models omit the intended visual semantics of whitespaces on documents. The second isolates a Kandinsky-specific suppression: the generated images consistently and more strongly than the natural-image baseline, indicating an overactive prior shared across the evaluated models.

such as Kandinsky’s omission of pan. The practical significance is that a single concept-energy comparison can surface surprising limitations that would be difficult to prioritize from aggregate image-quality metrics alone.

The distribution-level evidence also raises a finer-grained question: are blindspots only population-level shifts, or do the same energy differences reveal interpretable behavior at the individual datapoint level? We therefore next examine matched image pairs to separate prompt ambiguity, concept omission, and memorization-like alignment.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we rank individual natural-versus-generated image pairs by the average $\delta(\cdot)$ difference across all concepts and inspect the lowest- and highest-distance cases. This ranking makes the metric useful for interpreting both apparent successes and failures, with failures often traceable to prompt ambiguity and low-distance successes sometimes traceable to memorization artifacts. Fig. 11a shows examples with near-zero average $\delta(\cdot)$ differences. By the metric, these generated images are conceptually very close to the originals. Qualitative inspection, however, suggests that the low distance can arise from replication of memorized templates rather than faithful prompt understanding: repeated visual structures such as clothing outlines or object arrangements indicate copying from frequent training patterns. Fig. 11b shows the opposite regime, with samples among the largest $\delta(\cdot)$ differences. These pairs exhibit substantial concept divergence between the generated and natural images. Some divergences reflect underspecified or noisy captions, but others are clear blindspots: the prompt describes a concept visible in $\mathcal{D}_X$, yet the model does not realize that concept in $\mathcal{D}'_X$. Thus, the datapoint-level metric distinguishes practical failure modes even when the comparison is performed in the same shared concept space as the aggregate analysis.



Figure 11: **• Datapoint-level Conceptual Alignment.** (a) Matched pairs with minimal concept-energy differences, where the low metric distance coincides with apparent memorization of training patterns. (b) Matched pairs with large concept-energy differences, where prompt ambiguity or model limitations produce visible concept divergence.

384 5.5 Analyzing Post-Training Effects

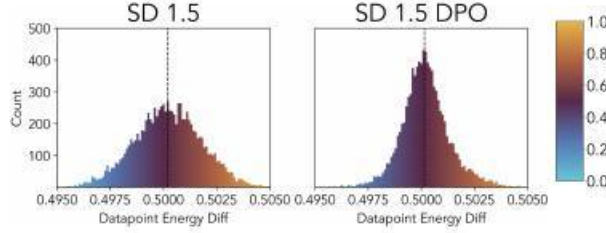


Figure 12: **Effect of DPO on Concept Fidelity.** Datapoint-wise histograms compare concept-energy distances between synthesized and natural images for SD 1.5 with and without DPO, using the same ℓ_2 distance between $\xi(\mathcal{D}'_{\mathcal{X}})$ and $\xi(\mathcal{D}_{\mathcal{X}})$.

385 Post-training protocols such as safety fine-tuning have been argued to reduce generation diver-
 386 sity [140]. Because our pipeline measures concept-energy distance to the natural baseline, we can use
 387 it to compare how post-training changes the model’s empirical concept distribution. We focus on DPO,
 388 a common safety fine-tuning protocol [141]. Specifically, we compare two SD 1.5 variants under the
 389 same evaluation condition: one trained with DPO and one without. For each matched image pair
 390 $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}'_{\mathcal{X}})$, we compute the ℓ_2 distance between their concept-energy vectors, $\|\xi(\mathcal{D}'_{\mathcal{X}}) - \xi(\mathcal{D}_{\mathcal{X}})\|_2$.
 391 Fig. 12 plots the resulting datapoint-wise distance distributions. The DPO-enhanced model has
 392 both a lower median and a narrower spread, indicating that generated concepts more consistently
 393 match the natural-image baseline. This pattern suggests that DPO regularizes the concept distribution
 394 by reducing large deviations from the semantic content of the paired natural images. Although
 395 this analysis does not isolate the precise inductive biases introduced by DPO, it provides empirical
 396 evidence that the human-preference objective improves alignment with the training distribution under
 397 our concept-energy metric. In particular, the comparison indicates reductions in *both over and under*
 398 activation of individual concepts relative to the natural baseline. These results illustrate a practical
 399 use of the pipeline: post-training interventions can be compared not only by perceived output quality,
 400 but also by how much they reshape the conceptual geometry of generated images.

401 5.6 Conceptual Misalignment as a Function of Empirical Frequency

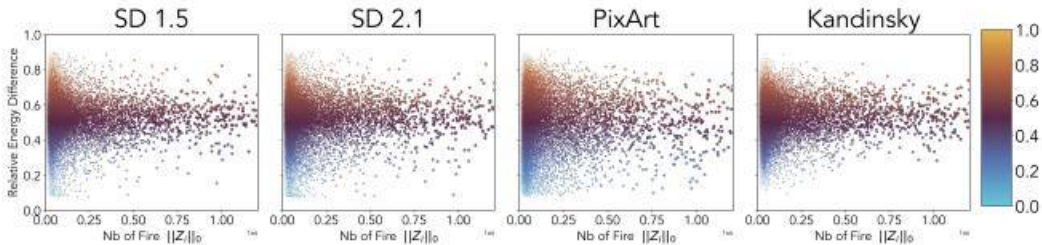


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots compare each concept’s empirical natural-image frequency (x-axis) with its spectrum difference (y-axis) for the four evaluated models. Each point is one concept, and point size is proportional to activation frequency, highlighting how long-tail concepts relate to larger prevalence margins.

402 Finally, we test whether the observed blindspots are explained only by architecture or also by
 403 empirical frequency in the data distribution. The hypothesis is that concepts activated rarely in
 404 natural images—the long tail of the distribution—should be more prone to large real-versus-generated
 405 prevalence margins. We process $\mathcal{D}_{\mathcal{X}}$ with the trained SAE and compute each concept’s empirical
 406 activation frequency $\|Z_{:,i}\|_0$, where $Z_{:,i}$ contains the activations of concept i across all images.
 407 We then compare this frequency to the absolute energy difference measured on generated outputs.
 408 Fig. 13 reports this relationship for all evaluated models. Concepts with higher natural frequency
 409 tend to have lower energy discrepancies, whereas rare concepts—especially suppressed concepts
 410 ($\delta(k) < 0.5$)—show larger alignment errors. This pattern supports a distributional interpretation
 411 of many blindspots: they are not random quirks, but systematic failures concentrated in long-tail

regions of the concept distribution. Practically, this points to remedies such as data reweighting or augmentation in addition to architectural changes.

6 Discussion

Across four popular generative image models, our results show that conceptual blindspots are measurable, structured, and practically interpretable. The same RA-SAE concept basis and energy-difference metric reveal heavy-tailed concept-prevalence margins, consistently stronger suppression than exaggeration, high agreement between related SD variants, weaker agreement with PixArt and Kandinsky, datapoint-level memorization artifacts, post-training shifts under DPO, and larger errors for long-tail concepts. Each pattern could support a dedicated study; here, we emphasize the breadth of evidence enabled by a single comparison framework and exploratory tool. Future work can extend the analysis to more architectures, larger evaluation sets, causal studies of blindspot origins, and hierarchical RA-SAE representations that separate coarse concepts from finer compositional variants.

Limitations. The quantitative evidence should be interpreted relative to the representation and sample used to obtain it. Because concept extraction relies on DINOv2 and RA-SAE, the measured margins cover only concepts that this pipeline represents well; concepts poorly captured by either component may be missed. Likewise, the 10,000-image evaluation set is large enough to reveal consistent empirical patterns across models, but it may still under-sample the rarest long-tail concepts, uncommon co-occurrences, and more complex compositional statistics.

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Appendix

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A Experimental Setup

This section specifies the evaluation protocol used to make quantitative comparisons across four popular generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. The code is available at a software repository.

A.1 Observation Space

The observation space contains 10,000 sampled image-text pairs from LAION-5B [93], which serves as the natural-image baseline for all real-versus-generated comparisons. Because the original release is no longer available due to CSAM and other unsafe-content concerns, we use the additionally filtered substitute subset at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure is: (1) load LAION with the Hugging Face datasets library, (2) validate URL structure and resource availability with HTTP HEAD requests, and (3) sample with replacement until 10,000 valid image-text pairs are obtained. The resulting D_G of (x, t) tuples provides an accessible and fixed natural baseline, which is necessary for comparing generated outputs when many original image URLs are no longer available. Additional matched synthesized examples appear in Appendix H

A.2 Synthesized Images

For each evaluated model, we generate a synthetic dataset D_{go} in one-to-one correspondence with D_G , producing matched triplets (x, x', t) . Thus, every model is evaluated on the same 10,000 captions and compared against the same natural images, making concept-energy differences attributable to the generator rather than to changes in prompts or baseline data.

Synthesis follows the standard text-to-image pipeline for each architecture using the Hugging Face diffusers library, with models loaded at mixed precision (fp16). All generated images use 512×512 resolution and default parameters, so model comparisons share the same output size and implementation stack.

Stable Diffusion 1.5. We use the checkpoint at <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5>, a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>. Inference uses 50 steps and guidance scale 7.5, establishing the SD 1.5 baseline for later comparisons.

Stable Diffusion 1.5 + DPO. The DPO variant used in Sec. 5.5 keeps the SD 1.5 implementation fixed and replaces only the UNet with the DPO-trained version at <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>, isolating the post-training comparison.

Stable Diffusion 2.1. We use <https://huggingface.co/stabilityai/stable-diffusion-2-1>. Inference uses 50 steps and guidance scale 7.5, matching the SD 1.5 sampling configuration where possible.

Kandinsky. We use <https://huggingface.co/kandinsky-community/kandinsky-2-1>. Inference uses 100 steps and guidance scale 4.0, following the model’s standard generation settings for the shared prompt set.

PixArt. We use <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS>. Inference uses 50 steps and guidance scale 7.5, placing PixArt under the same matched-prompt evaluation protocol.

A.3 Distribution Level Analysis

Section 5.1 For every evaluated model, we compute $\delta(\cdot)$ over all 32,000 concepts and normalize with a sigmoid transformation at temperature $T = 0.8$. We then visualize the values as log-scale density histograms with 100 bins over $[0, 1]$, making the suppressed and exaggerated tails directly comparable across models.

904 **Section 5.2** We embed all 32,000 sparse concept codes into two dimensions with UMAP. Each
 905 point is one concept colored by $\delta(\cdot)$, so spatial clusters show where suppressed and exaggerated
 906 margins concentrate. Cross-model consistency is quantified with pairwise Pearson correlations
 907 between full $\delta(\cdot)$ vectors, reported as scatter plots and correlation matrices, thereby separating
 908 model-specific blindspots from universal patterns.

909 **Section 5.3** We rank all 32,000 concepts by $\delta(\cdot)$ and inspect both extremes of the distribution.
 910 Suppressed examples are selected from concepts with $\delta(\cdot) < 0.1$, while exaggerated examples are
 911 selected from the high- $\delta(\cdot)$ regime. The displayed concepts are manually annotated by inspecting
 912 their most activating images and spatial attention patterns, and Appendix D describes ongoing work
 913 toward automating this interpretation step.

914 **Section 5.5** We compare SD 1.5 with and without DPO by computing, for each matched pair
 915 $(\mathbf{x}, \mathbf{x}')$, the L2 distance between concept-energy vectors $\|\xi(\mathbf{x}') - \xi(\mathbf{x})\|_2$. Element-wise differences
 916 are sigmoid-normalized with temperature $T = 0.8$ before averaging, yielding a datapoint-wise
 917 concept distance. Overlapping histograms then compare the two variants under the same metric and
 918 prompt set.

919 **Section 5.6** For each concept c_k , we compute empirical frequency as $\|\mathbf{Z}_{:,i}\|_0$, the number of non-
 920 zero activations in the natural dataset. We apply sigmoid normalization with temperature $T = 0.4$
 921 to $\delta(\cdot)$ and visualize frequency-versus-margin scatter plots. The x-axis reports empirical concept
 922 frequency, the y-axis reports sigmoid-transformed energy difference, point size is proportional to
 923 activation frequency, and color encodes the magnitude of the energy difference.

924 **A.4 • Datapoint Level Analysis**

925 **Section 5.4** For each matched image pair $(\mathbf{x}, \mathbf{x}')$, we compute $\|\xi(\mathbf{x}') - \xi(\mathbf{x})\|_2$ as a scalar concept-
 926 space distance. Ranking pairs by this distance highlights both low-divergence cases, which can reveal
 927 memorization artifacts, and high-divergence cases, which point to substantial conceptual failures.
 928 This converts datapoint inspection into a metric-guided comparison.

929 **A.5 • Co-occurrence Analysis**

930 **Appendix F** For the natural and synthesized datasets D_G and D_{go} , we analyze concept co-
 931 occurrence through the co-activation matrix $Z^T Z$, whose entries summarize pairwise concept usage.
 932 Eigendecomposition identifies dominant conceptual directions, and cosine-similarity heatmaps be-
 933 tween the top-100 eigenvectors compare natural and generated co-occurrence structure. In these
 934 100×100 matrices, diagonal alignment indicates shared principal concept axes, whereas off-diagonal
 935 structure indicates rotations or mismatches in compositional geometry.

936 B Custom RA-SAE Training Details

937 This section reports the training details for the custom archetypal SAE (RA-SAE) [56] that provides
938 the shared 32,000-concept measurement basis used throughout the paper. The model is open-sourced
939 at a software repository.

940 **Dataset.** The auto-encoder is trained on the complete ImageNet-1k training split, containing
941 approximately 1.28 M RGB images. Each image is converted into 261 DINOv2 visual tokens [135],
942 which are passed to the SAE without class or position embeddings. Across 50 epochs, this yields
943 $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$ training tokens for the concept dictionary.

944 **Dictionary.** The dictionary contains 32,000 concept dimensions. Sparsity is enforced with top- k
945 masking at $k = 5$, so only the five largest activations per input remain non-zero. We initialize weights
946 with Xavier/Glorot, train in mixed precision (fp16), and run the final ten epochs at full precision to
947 stabilize the learned measurement basis.

948 **Optimizer and Schedule.** We train for 50 epochs with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and
949 weight decay 10^{-5} . The learning rate linearly warms up for the first 5% of steps and then follows
950 cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The loss combines MSE reconstruction with an
951 auxiliary penalty for activations that never enter the top- k set, using $\lambda = 10^{-5}$.

952 C Computational Resources

953 This section summarizes the compute required to produce the reported quantitative comparisons. In
954 total, training, generation, and analysis used approximately 202 GPU-hours on NVIDIA H100s and
955 H200s.

956 **RA-SAE.** Training the shared concept extractor required approximately 24 GPU-hours on three
957 NVIDIA H100s.

958 **Synthesized Images.** Generating each full matched $D_{g\theta}$ (Appendix A) took roughly 5 hours per
959 generator when distributed across four NVIDIA H200 GPUs. Across five generators, this totaled
960 approximately 100 NVIDIA H200 GPU-hours.

961 **● Distribution-Level Analysis.** Computing distribution-level energy differences took about 3
962 hours per generator on a single NVIDIA H200 GPU, totaling approximately 15 GPU-hours.

963 **● Datapoint-Level Analysis.** Computing datapoint-level distances, ranking concepts per datapoint,
964 and ranking datapoints per concept also required about 3 hours per generator on one H200 GPU, or
965 roughly 15 GPU-hours total.

D Autointerpretability

To reduce manual effort in interpreting RA-SAE concepts, we qualitatively tested whether Vision-Language Models (VLMs) can describe concepts selected by the quantitative pipeline. We queried ChatGPT 4o through the OpenAI API with multiple prompts and visual formats for an image x'' in which a target concept was salient, then recorded the generated description. For this exploratory evaluation we allowed full reasoning; a scalable deployment could request only the final concept description.

First, we provided two inputs for the same concept: the raw image x'' and a heatmap-overlay version intended to localize where the RA-SAE activation was strongest:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DiNOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DiNOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

In this condition, the VLM did not align its description with the RA-SAE evidence. It described the foreground ring rather than the background wood texture, even though the heatmap indicated that the wood region carried the concept activation.

We then tested whether the mismatch was caused by prompt underspecification by repeating the comparison with a longer prompt that explained the heatmap more explicitly:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DiNOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DiNOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

The expanded prompt did not improve alignment: the VLM again selected the ring as the prominent concept. Additional variants, including prompts that described the heatmap generation procedure in more detail, produced the same qualitative outcome.

983 We next removed the raw image and gave the VLM only the heatmap-superimposed x'' with a similar
 984 explanation of how to interpret the overlay:

Input	Output
 You are given an image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.	The concept identified by the DINOv2-based SAE in this image appears to be "ring" or more broadly, "circular metallic object". The most salient (red) areas are focused tightly around the ring itself—especially along its edges and reflective interior—indicating that the model has strongly associated those regions with the concept. The surrounding wooden surface is mostly in blue and green, which shows it is not relevant to the concept, supporting the idea that the model is isolating the ring as the key object of interest.

985 This format also failed to shift the description toward the wooden background; the VLM continued
 986 to focus on the ring. Additional prompts with the same image format produced largely unchanged
 987 results.

988 These trials suggest that heatmaps alone are insufficient evidence for steering the VLM toward the
 989 RA-SAE-salient region when visually salient but irrelevant objects remain visible. We therefore
 990 replaced the heatmap overlay with an alpha-masked version of x'' that suppresses low-activation
 991 regions:

Input	Output
 You are given an image where only areas that contain a concept identified by a DINOv2-based SAE are preserved; areas without the concept were removed. Describe what is this concept.	The concept identified and preserved in this image appears to be "wood texture" or "wood grain." The retained areas show consistent patterns typical of wooden surfaces—linear, fibrous streaks in earthy tones like beige, gray, and brown. The selective masking highlights parts of the image where this natural texture is most prominent, suggesting the SAE learned to isolate the visual features of wood (likely flooring or a table).

992 With alpha masking, the VLM correctly identified wood texture as the dominant concept in the
 993 example. Qualitatively, this format was more consistent across the concepts and images we inspected.

994 The alpha-masked formatting and associated prompt produced the most reliable qualitative interpreta-
 995 tions in our exploration. The evidence remains preliminary because we lack ground-truth concept
 996 labels and evaluate only one VLM, but it provides a practical route for scaling annotation of the
 997 strongest quantitative blindspot candidates.

998 **E Additional Results: Model-Specific Blindspots**

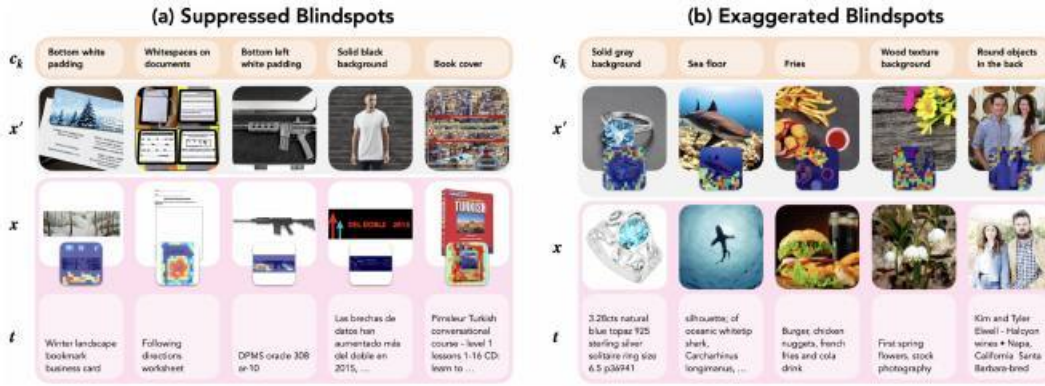


Figure 14: Conceptual blindspot examples for **Stable Diffusion 1.5**. For each measured concept, we show the highest-activation natural image for suppressed blindspots or generated image for exaggerated blindspots, with the concept heatmap overlaid to indicate the evidence supporting the comparison.

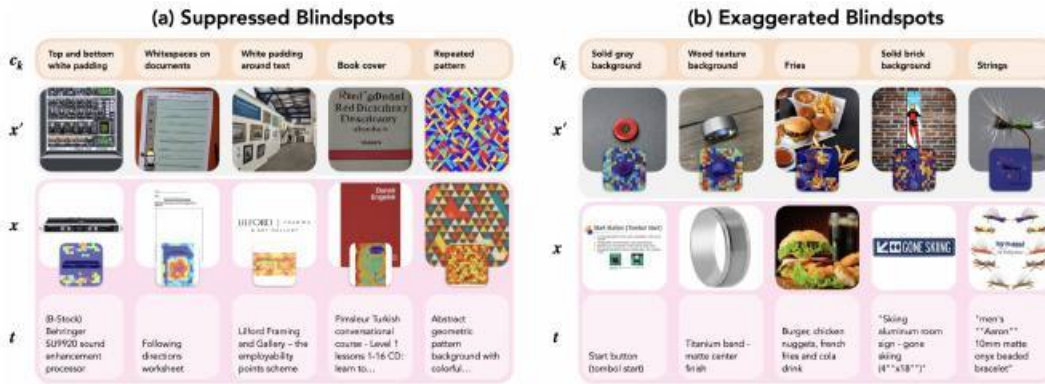


Figure 15: Conceptual blindspot examples for **Stable Diffusion 2.1**. Each panel uses the highest-activation natural image for suppressed concepts or generated image for exaggerated concepts, and the overlaid heatmap localizes the concept evidence used in the energy-difference ranking.

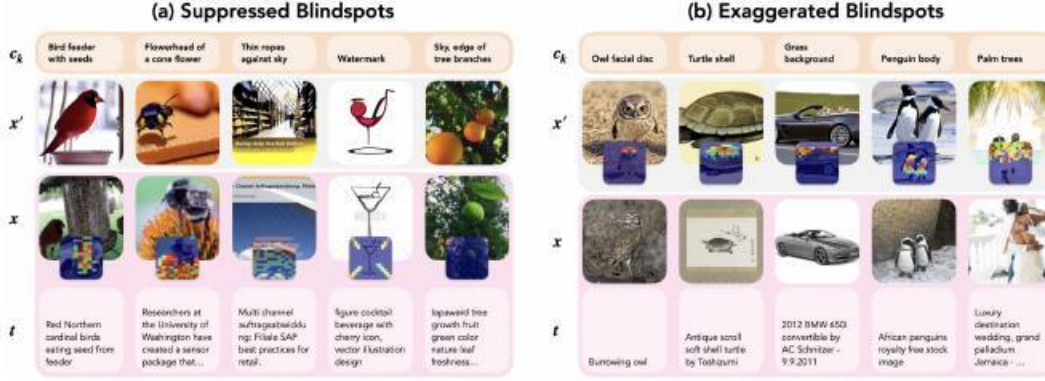


Figure 16: Conceptual blindspot examples for **Kandinsky**. The displayed image is the highest-activation natural exemplar for suppressed blindspots or generated exemplar for exaggerated blindspots, with the spatial heatmap showing where the RA-SAE concept is expressed.



Figure 17: Conceptual blindspot examples for **PixArt**. For each concept, the panel shows the highest-activation natural exemplar for suppression or generated exemplar for exaggeration, with heatmaps marking the image regions that drive the measured concept activation.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

The main analysis focuses on marginal concept activations, but visual scenes also depend on higher-order co-occurrence structure. We therefore compare natural and generated images through the co-activation matrix $Z^T Z$, whose entries measure pairwise concept usage and provide a quantitative view of compositional semantics.

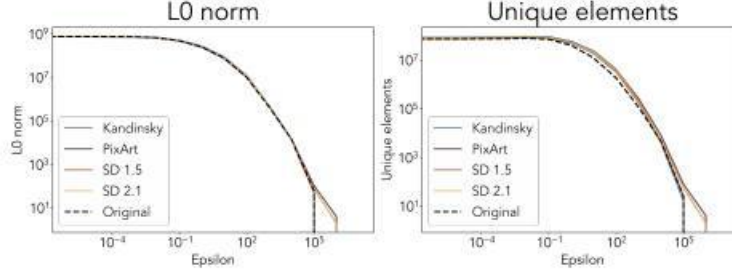


Figure 18: **Sparsity and Structural Divergence.** Left: the L0 norm of the co-occurrence matrix $Z Z^T$ as a function of threshold ϵ , measuring how many pairwise concept entries remain active for each model. Right: the number of synthesized-distribution entries that are unique relative to the natural baseline. The comparison shows that models preserve global sparsity while diverging in which pairings activate.

At the level of binary structure, diffusion models preserve the global sparsity of the natural co-occurrence matrix with high fidelity. In Fig. 18 (left), the ℓ_0 norm of thresholded $Z^T Z$ tracks closely between natural and synthesized distributions across all models over varying ϵ . This indicates that coarse graph connectivity—which concepts co-activate at all—is largely retained. Formally, $Z^T Z$ can be viewed as the adjacency matrix of a weighted, undirected concept graph, with edge weights given by dataset-level co-activation strength.

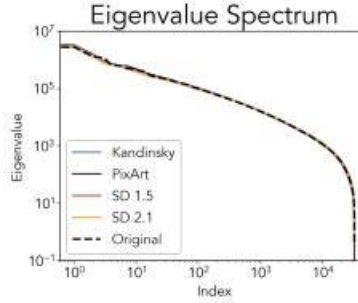


Figure 19: **Spectral Structure of Co-occurrence.** Log-log eigenvalue spectra of the co-occurrence matrices $Z Z^T$ compare each model against the natural distribution. All evaluated models match the natural distribution’s heavy-tailed decay, indicating similar global variance allocation across concept directions.

The content of those co-activations, however, differs from the natural baseline. Fig. 18 (right) shows that a substantial portion of generated $Z^T Z$ entries are unique relative to the natural distribution. Thus, the models preserve the overall capacity for compositionality while shifting which concept pairs are active, a pattern consistent with model-specific inductive biases or training artifacts. We next quantify the internal structure of these co-occurrence matrices spectrally. Fig. 19 compares eigenvalue spectra for each model and the natural distribution. The heavy-tailed decay appears in all spectra, consistent with power-law behavior and indicating that generative models preserve broad rank structure and variance allocation across conceptual dimensions.

Finally, we compare dominant concept directions directly using cosine-similarity heatmaps between the top 100 eigenvectors of synthesized and natural co-occurrence matrices (Fig. 20). All models

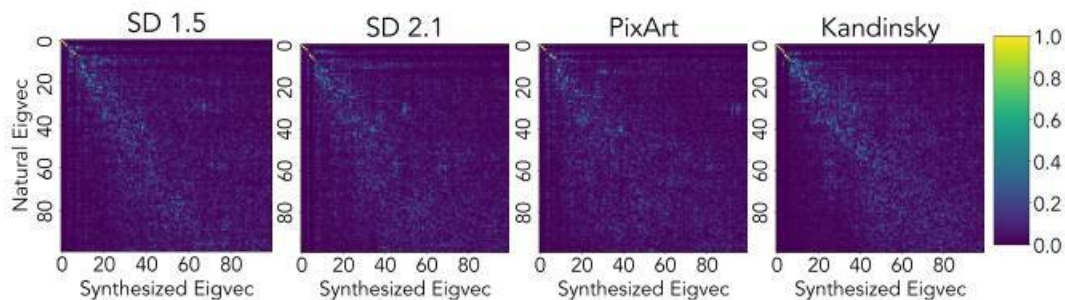


Figure 20: ● **Concept Basis Similarity.** Cosine-similarity heatmaps compare the top 100 eigenvectors of natural and synthesized co-occurrence matrices ZZ^T . Strong diagonal structure indicates aligned dominant concept directions, while off-diagonal mass quantifies rotations and mismatches that differ across models.

1022 show partial diagonal alignment, meaning their leading concept axes overlap with the natural baseline,
 1023 but off-diagonal entries reveal rotations and higher-mode mismatches in compositional geometry.

1024 Together, these comparisons show that diffusion models match several global co-activation statistics
 1025 while still deviating in the identity and orientation of higher-order concept structure. Such structured
 1026 residual differences may help explain practical failures in coherent multi-object scenes or relational
 1027 concepts.

1028 G Additional Results: Blindspot Stress Testing

1029 To stress-test the measured blindspots, we generated additional prompts explicitly targeting selected
 1030 concepts and compared models where the concept was identified as a blindspot against models
 1031 where it was not. This provides an out-of-sample qualitative check on the original energy-difference
 1032 rankings.

1033 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1034 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1035 *text-to-image model. Return these as a list of strings in Python.* For each prompt, we generated
 1036 five images and analyzed them with the custom RA-SAE model (Appendix B), ranking outputs by
 1037 the intensity of the target concept. We then manually reviewed all images to determine whether the
 1038 concept was successfully depicted.

1039 In the suppressed-concept examples below, models sometimes generate partial evidence of the
 1040 target concept, such as a holder or string for bird feeder or a round hole for glossy DVD disc.
 1041 However, they generally fail to synthesize the complete concept, matching the original blindspot
 1042 classification and supporting the stress-test comparison.

1043 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Stress-test generations for prompts targeting bird feeder. Kandinsky was identified by our metric as suppressing this concept, and the generated images fail to depict the full object. SD 1.5 was not identified as having this blindspot and produces clearer bird-feeder examples under the same targeted prompt condition.



Figure 22: Stress-test generations for prompts targeting glossy DVD disc. SD 1.5 was identified by our metric as suppressing this concept, and its outputs do not recover the complete target object. Kandinsky was not identified as having this blindspot and generates clearer examples under the same targeted prompt condition.

1045 **H Additional Examples: Synthesized Images**



Figure 23: Additional matched evaluation examples. Each LAION-5B image-caption pair is shown with images generated from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky, illustrating the shared comparison condition used throughout the quantitative analysis.

1046 I Concentration bounds for δ

1047 Our experiments estimate $\delta(k)$ from $n = 10,000$ paired samples per concept. To justify this sample
 1048 size for concept-level comparisons, we derive a concentration bound for the empirical estimator $\hat{\delta}_n(k)$
 1049 using McDiarmid’s inequality [142]. The bound shows that the reported margins can be estimated
 1050 reliably even with this finite evaluation budget.

1051 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1052 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1053 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1054 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1055 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1056 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1057 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1058 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1059 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1060 which proves the claim. □

1061 In practice, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only a
 1062 few reaching values up to 10. The concentration bound indicates that, even for the largest observed
 1063 biases, $n = 10,000$ paired samples yield $\hat{\delta}_n(k)$ estimates that remain within a small ε of the true
 1064 value with high probability. This supports the sampling strategy and the reliability of the reported
 1065 concept-bias measurements.

1066 J Monotonicity and Calibration-Free Interpretation of δ

1067 For blindspot analysis, the key operation is ranking concepts by the severity of their generated-versus-
 1068 natural prevalence margin. We use $\delta(k)$ for this purpose, so we verify that this bounded score does
 1069 not distort rankings relative to the raw energy gap or odds ratio. The result below shows that $\delta(k)$ is a
 1070 strictly increasing reparameterization of both quantities, so no additional calibration is needed for
 1071 ranking concepts.

1072 **Theorem 5** (Monotonicity and Calibration of $\delta_{g\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1073 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_X}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_X}[\xi_k(\mathbf{x})],$$

1074 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1075 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1076 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1077 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1078 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1079 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1080 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1081 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1082 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1083 concepts.

1084

□

1085 Therefore, ranking concepts by $\delta(\cdot)$ is exactly equivalent to ranking them by energy gap or conceptual
 1086 generation odds ρ_k . The empirical comparisons can use the bounded δ scale without changing the
 1087 ordering of blindspot severity.

K Stability of FID Under SAE Embeddings

This section establishes a quantitative relationship between Fréchet Inception Distance (FID) in the original d -dimensional activation space and FID after applying a potentially overcomplete SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout, we assume D has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We begin with the simple non-overcomplete case: if D is orthogonal and $k = d$, the map is an isometry between A and Z , so FID is exactly preserved. We then consider the realistic overcomplete case and show that FID distortion is bounded by the extreme singular values of D . The argument uses the Wasserstein-2 metric $\mathcal{W}_2$, with FID interpreted as $\mathcal{W}_2^2$ specialized to Gaussian surrogates.

For a probability measure μ on $\mathbb{R}^d$, write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Let $\sigma_{\min}$ and $\sigma_{\max}$ be the minimal and maximal singular values of D , equivalently the square roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically, $\sigma_{\min}, \sigma_{\max} \approx 1$ is often observed, but the proof does not rely on this favorable case. We start with a simple lemma for non-overcomplete D .

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D preserves Euclidean distances. Since the push-forward applies the same isometric map inside the $\mathcal{W}_2$ infimum, the transport integrand and therefore the infimum value are unchanged. $\square$

This exact-isometry case is not the typical SAE setting, since SAEs usually rely on overcompleteness to extract meaningful and interpretable concepts. In the overcomplete case, D is not square orthonormal, but a column-orthonormal dictionary still yields a controlled distortion bound through the extremal singular values of $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Let μ denote the empirical measure of A and ν the empirical measure of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating this bound with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they also hold for the optimal couplings defining $\mathcal{W}_2(\mu, \nu)$ and $\mathcal{W}_2(\mu_D, \nu_D)$, even though the optima need not coincide.

1122 Taking the infimum over π term-wise makes the comparison explicit:

$$\sigma_{\min}^2 \inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|_2^2 d\pi \leq \inf_{\pi \in \Pi(\mu, \nu)} \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|_2^2 d\pi.$$

1123 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\mu, \nu) \leq \mathcal{W}_2^2(\mu_D, \nu_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\mu, \nu).$$

1124 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and substituting (A, A') and (Z, Z')
 1125 gives the stated bound. $\square$

1126 Theorem 6 therefore frames SAE-space FID as a controlled comparison: a column-orthogonal
 1127 overcomplete dictionary can distort Fréchet Inception Distance only by the square of its extremal
 1128 singular values. When D is nearly orthogonal—as suggested empirically by $\sigma_{\min}, \sigma_{\max} \approx 1$ —FID
 1129 measured in the SAE feature space remains close to canonical FID. $\square$

1130